

2025-10-24 - Servidio (8)

Previous lecture: [2025-10-23 - Servidio \(7\)](#)

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From the 2 fluids equations

$$\left\{ \begin{array}{l} \frac{\partial n_\alpha}{\partial t} + \nabla \cdot (n_\alpha \bar{U}_\alpha) = 0 \\ \frac{\partial m_\alpha n_\alpha \bar{U}_\alpha}{\partial t} + \nabla \cdot (m_\alpha n_\alpha \bar{U}_\alpha \bar{U}_\alpha) + \nabla P^\alpha - q_\alpha n_\alpha [\bar{E} + \bar{v} \times \bar{B}] = 0 \\ P^\alpha = A n_\alpha^\gamma \\ \nabla \cdot \bar{E} = \frac{1}{\varepsilon_0} \sum_\alpha q_\alpha n_\alpha \\ \frac{\partial \bar{B}}{\partial t} = -\nabla \times \bar{E} \\ \nabla \times \bar{B} = \mu_0 \bar{J} + \mu_0 \varepsilon_0 \frac{\partial \bar{E}}{\partial t} \\ \nabla \cdot \bar{B} = 0 \end{array} \right. \quad \begin{array}{l} (A) \\ (B) \\ (C) \\ (D) \\ (E) \\ (F) \\ (G) \end{array}$$

We want a simpler version of this system

Compressible Magnetohydrodynamics (MHD)

Protons in a magnetic field rotate at their gyro-cyclotron frequency

$$\omega_{cp} = \frac{eB}{m_p c}$$

Eg. solar coronal loops can be studied at the scale of hours or days.

We have

$$\omega \ll \omega_{cp}$$

The ion skin depth is

$$d_p = \frac{c}{\omega_p}, \quad \omega_p = \sqrt{\frac{n_p e^2}{m_p \varepsilon_0}}$$

On the Sun $d_p \simeq 2 \text{ m}$.

We want

$$L_0 \gg d_p$$

with L_0 the typical size of a coronal mass ejection.

We apply the following simplifications:

$$m_p \gg m_e; \quad n_p \sim n_e = n$$

the quasi-neutrality condition is also justified by the fact that any dishomogeneities in n_p or n_e is quickly restored by the small scale electric field using high frequencies.

Continuity equations

So

$$\rho_c = e(n_p - n_e) = 0$$

and

$$\rho = \sum_{\alpha} m_{\alpha} n_{\alpha} = m_p n_p + n_e m_e = n(m_p + m_e) \simeq n m_p$$

$$\rho \bar{U} = m_p n_p \bar{U}_p + m_e n_e \bar{U}_e \simeq \rho \bar{U}_p$$

So $\rho \sim n m_p$ and $\bar{U} \sim \bar{U}_p$.

Now we use the two variables ρ and $\bar{U}$, we multiply the continuity equation (A) for a factor m_{α}

$$\begin{cases} \frac{\partial n_p}{\partial t} + \nabla \cdot (n_p \bar{U}_p) = 0 \\ \frac{\partial n_e}{\partial t} + \nabla \cdot (n_e \bar{U}_e) = 0 \end{cases} \implies \begin{cases} \frac{\partial m_p n_p}{\partial t} + \nabla \cdot (m_p n_p \bar{U}_p) = 0 \\ \frac{\partial m_e n_e}{\partial t} + \nabla \cdot (m_e n_e \bar{U}_e) = 0 \end{cases}$$

We sum them and we obtain

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{U}) = 0$$

Euler equations (sum)

For the moment equation (B) first we sum the equation for protons and the one for electrons

$$\begin{aligned} & \frac{\partial \left[n \left(m_p \bar{U}_p + \cancel{m_e \bar{U}_e} \right) \right]}{\partial t} + \nabla \cdot \left[n \left(m_p \bar{U}_p \bar{U}_p + \cancel{m_e \bar{U}_e \bar{U}_e} \right) \right] + \\ & + \nabla (P_p + P_e) + en \left[\bar{U}_e \times \bar{B} - \bar{U}_p \times \bar{B} \right] = 0 \\ \implies & \frac{\partial n m_p \bar{U}}{\partial t} + \nabla \cdot (n m_p \bar{U} \bar{U}) + \nabla (P_{tot}) + en (\bar{U}_e - \bar{U}_p) \times \bar{B} = 0 \\ \implies & \frac{\partial \rho \bar{U}}{\partial t} + \nabla \cdot (\rho \bar{U} \bar{U}) + \nabla (P_{tot}) - \bar{J} \times \bar{B} = 0 \end{aligned}$$

Euler equations (difference)

Now we multiply equation (B) by the factor $|q_{\alpha}|/m_{\alpha}$

$$\begin{cases} \frac{\partial (en_p \bar{U}_p)}{\partial t} + \nabla \cdot [en_p \bar{U}_p \bar{U}_p] + \frac{e}{m_p} \nabla P_p - \frac{e^2}{m_p} n_p [\bar{E} + \bar{U}_p \times \bar{B}] = 0 & (1) \end{cases}$$

$$\begin{cases} \frac{\partial (en_e \bar{U}_e)}{\partial t} + \nabla \cdot [en_e \bar{U}_e \bar{U}_e] + \frac{e}{m_e} \nabla P_e - \frac{e^2}{m_e} n_e [\bar{E} + \bar{U}_e \times \bar{B}] = 0 & (2) \end{cases}$$

We evaluate the first equation minus the second ((1) - (2))

$$\frac{\partial \bar{J}}{\partial t} + \nabla \cdot \left[en \left(\bar{U}_p \bar{U}_p - \bar{U}_e \bar{U}_e \right) \right] + \frac{e}{m_p} \nabla P_p - \frac{e}{m_e} \nabla P_e +$$

$$- \frac{e^2}{m_p} n \left[\bar{E} + \bar{U}_p \times \bar{B} \right] - \frac{e^2}{m_e} n \left[\bar{E} + \bar{U}_e \times \bar{B} \right] = 0$$

We multiply by m_e and approximate $m_e \rightarrow 0$, so only terms $\sim 1/m_e$ survive

$$- \cancel{\nabla P_e} - e^2 n \left[\bar{E} + \bar{U}_e \times \bar{B} \right] = 0$$

$$\implies \bar{E} = -\bar{U}_e \times \bar{B} - \frac{1}{n_e e} \nabla P_e$$

In which we use

$$\bar{J} = en\bar{U}_p - en\bar{U}_e \implies \bar{U}_e = \bar{U}_p - \frac{\bar{J}}{n_e e}$$

so

$$\bar{E} = -\bar{U} \times \bar{B} + \frac{1}{n_e e} \bar{J} \times \bar{B} - \frac{1}{n_e e} \nabla P_e$$

Ampère-Maxwell law

From Ampère-Maxwell law (F) we have

$$\nabla \times \bar{B} = \mu_0 \bar{J} + \frac{1}{c} \frac{\partial \bar{E}}{\partial t}$$

We neglect the second term because we are not in the relativistic regime (we are studying low frequencies), so

$$\bar{J} = \frac{1}{\mu_0} \nabla \times \bar{B}$$

Obtaining the Ideal Ohm's law

We use this result for $\bar{J}$ in the expression for $\bar{E}$

$$\bar{E} = -\bar{U} \times \bar{B} + \frac{1}{\mu_0 m_e} \left(\nabla \times \bar{B} \right) \times \bar{B} - \frac{1}{n_e e} \nabla P_e$$

Now we divide by the typical length of the system L_0 and we obtain

$$\nabla = L_0^{-1} \tilde{\nabla}, \quad \bar{U} = U_0 \tilde{U}$$

$$\bar{B} = B_0 \tilde{B}, \quad P = P_0 \tilde{P}$$

$$n = n_0 \tilde{n}$$

So

$$\frac{\tilde{E} E_0}{L_0} = -\bar{U} \times \bar{B} \frac{U_0 B_0}{L_0} + \frac{B_0^2}{\mu_0 n_0 e L_0^2} \left(\tilde{\nabla} \times \tilde{B} \right) \times \tilde{B} - \frac{P_0}{n_0 e L_0^2} \tilde{\nabla} \tilde{P}_e$$

The last two terms are comparable to L_0^{-2} so they are negligible.

We finally obtain the ideal Ohm's law

$$\bar{E} = -\bar{U} \times \bar{B}$$

which is valid also in observation of the solar wind, in the heliosphere, in the magnetopause, ecc.

Summary of compressible MHD

$$\left\{ \begin{array}{l} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{U}) = 0 \\ \frac{\partial (\rho \bar{U})}{\partial t} + \nabla \cdot [\rho \bar{U} \bar{U}] + \nabla P_{tot} - \bar{J} \times \bar{B} = 0 \\ P = A \rho^\gamma \\ \frac{\partial \bar{B}}{\partial t} = -\nabla \times \bar{E} = \nabla \times (\bar{U} \times \bar{B}) \\ \nabla \cdot \bar{B} = 0 \end{array} \right.$$

with

$$\left\{ \begin{array}{l} \bar{E} = -\bar{U} \times \bar{B} \\ \bar{J} = \frac{1}{\mu_0} \nabla \times \bar{B} \end{array} \right.$$

We recall the Faraday's law (which is used for magnetic field on Earth)

$$\frac{\partial \bar{B}}{\partial t} = \nabla \times (\bar{U} \times \bar{B})$$

Incompressible Ideal Magnetohydrodynamics

The incompressibility condition is

$$U \ll c_s \quad \implies \quad \nabla \cdot \bar{U} = 0$$

with

$$c_s^2 = \frac{\partial P}{\partial \rho} \rightarrow \infty$$

We rewrite the **continuity equation** as

$$\begin{aligned}
& \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{U}) = 0 \\
\Rightarrow & \frac{\partial \rho}{\partial t} + \rho \nabla \cdot \bar{U} + \bar{U} \cdot \nabla \rho = 0 \\
\Rightarrow & \frac{\partial \rho}{\partial t} + \bar{U} \cdot \nabla \rho = 0 \\
\Rightarrow & \left(\frac{\partial}{\partial t} + \bar{U} \cdot \nabla \right) \rho = 0 \\
\Rightarrow & \frac{d}{dt} \rho = 0
\end{aligned}$$

where the last derivative is a total (Lagrangian) derivative.

From now on this will be trivial and ρ_0 will be the typical density.

The **Euler equation** is (using the continuity equation)

$$\begin{aligned}
& \frac{\partial (\rho \bar{U})}{\partial t} + \nabla \cdot [\rho \bar{U} \bar{U}] + \nabla P_{tot} - \bar{J} \times \bar{B} = 0 \\
\Rightarrow & \rho \frac{\partial \bar{U}}{\partial t} + \bar{U} \frac{\partial \rho}{\partial t} + \bar{U} \nabla \cdot (\rho \bar{U}) + \rho (\bar{U} \cdot \nabla) \bar{U} + \nabla P_{tot} - \bar{J} \times \bar{B} = 0 \\
\Rightarrow & \rho \frac{\partial \bar{U}}{\partial t} + \bar{U} \underbrace{\left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \bar{U}) \right]}_{=0} + \rho (\bar{U} \cdot \nabla) \bar{U} + \nabla P_{tot} - \bar{J} \times \bar{B} = 0 \\
\Rightarrow & \frac{\partial \bar{U}}{\partial t} + (\bar{U} \cdot \nabla) \bar{U} = -\frac{\nabla P_{tot}}{\rho} + \frac{\bar{J} \times \bar{B}}{\rho} \\
\Rightarrow & \frac{\partial \bar{U}}{\partial t} + (\bar{U} \cdot \nabla) \bar{U} = -\nabla \left(\frac{P_{tot}}{\rho_0} \right) + \frac{(\nabla \times \bar{B}) \times \bar{B}}{\mu_0 \rho_0}
\end{aligned}$$

And for the magnetic field (**Faraday's law**) we have

$$\begin{aligned}
\frac{\partial \bar{B}}{\partial t} &= \nabla \times (\bar{U} \times \bar{B}) = \\
&= -\bar{U} \cdot \nabla \bar{B} + \bar{B} \cdot \nabla \bar{U} - \cancel{\bar{U} \nabla \cdot \bar{B}} + \cancel{\bar{B} \nabla \cdot \bar{U}} = \\
&= -\bar{U} \cdot \nabla \bar{B} + \bar{B} \cdot \nabla \bar{U}
\end{aligned}$$

So we have

$$\begin{cases} \frac{\partial \bar{U}}{\partial t} + \bar{U} \cdot \nabla \bar{U} = -\nabla \left(\frac{P}{\rho_0} \right) + \frac{(\nabla \times \bar{B}) \times \bar{B}}{\mu_0 \rho_0} \\ \frac{\partial \bar{B}}{\partial t} = -\bar{U} \cdot \nabla \bar{B} + \bar{B} \cdot \nabla \bar{U} \end{cases}$$

with $\nabla \cdot \bar{U} = \nabla \cdot \bar{B} = 0$.

The **EoS in MHD**.

We leave the thermodynamics ($c_s \rightarrow \infty$) and the pressure comes from $\nabla \cdot \bar{U} = 0$.

We use

$$\frac{\partial}{\partial t} \nabla \cdot \bar{U} = 0$$

and from the divergence of Euler equation we have

$$\begin{aligned} \cancel{\frac{\partial(\nabla \cdot \bar{U})}{\partial t}} + \nabla \cdot [\bar{U} \cdot \nabla \bar{U}] &= -\nabla^2 \left(\frac{P}{\rho_0} \right) + \nabla \cdot \left(\frac{(\nabla \times \bar{B}) \times \bar{B}}{\mu_0 \rho_0} \right) \\ \implies \nabla^2 P &= -\rho_0 \nabla \cdot \left[\bar{U} \cdot \nabla \bar{U} - \frac{(\nabla \times \bar{B}) \times \bar{B}}{\mu_0 \rho_0} \right] \equiv G(\bar{x}, t) \end{aligned}$$

So we need to solve a laplace differential equation in order to find the pressure.

Alfvénic units

We reduce the number of parameters

$$\begin{cases} \frac{\partial \bar{U}}{\partial t} + \bar{U} \cdot \nabla \bar{U} = -\nabla \left(\frac{P}{\rho_0} \right) + \frac{(\nabla \times \bar{B}) \times \bar{B}}{\mu_0 \rho_0} \\ \frac{\partial \bar{B}}{\partial t} = \nabla \times (\bar{U} \times \bar{B}) \end{cases}$$

we define the adimensional quantities

$$\begin{aligned} x &= L_0 \tilde{x}, & \bar{U} &= U_0 \tilde{U} = \frac{L_0}{T_0} \tilde{U}, \\ t &= T_0 \tilde{t}, & P &= P_0 \tilde{P} = \rho_0 U_0^2 \tilde{P}, \\ \bar{B} &= B_0 \tilde{B} \end{aligned}$$

We normalize the Euler equation

$$\begin{aligned} \frac{U_0}{T_0} \frac{\partial \tilde{U}}{\partial \tilde{t}} + \frac{U_0^2}{L_0} \tilde{U} \cdot \tilde{\nabla} \tilde{U} &= -\frac{P_0}{\rho_0 L_0} \tilde{\nabla} \tilde{P} + \frac{B_0^2}{\rho_0 \mu_0 L_0} (\tilde{\nabla} \times \tilde{B}) \times \tilde{B} \\ \implies \frac{\partial \tilde{U}}{\partial \tilde{t}} + \tilde{U} \cdot \tilde{\nabla} \tilde{U} &= -\tilde{\nabla} \tilde{P} + \frac{B_0^2 T_0}{\rho_0 \mu_0 L_0 U_0} (\tilde{\nabla} \times \tilde{B}) \times \tilde{B} \end{aligned}$$

We define the Alfvén velocity as

$$U_A = \frac{B_0}{\sqrt{\rho_0 \mu_0}}$$

this velocity defines the Alfvénic plasma, where $U_0 = U_A$

$$\frac{B_0^2 T_0}{\rho_0 \mu_0 L_0 U_0} = \frac{B_0^2}{\rho_0 \mu_0} \frac{1}{U_0^2} = \frac{U_A^2}{U_0^2} = 1$$

Now we normalize the magnetic field equation

$$\frac{B_0}{T_0} \frac{\partial \tilde{\tilde{B}}}{\partial \tilde{t}} = \frac{U_0 B_0}{L_0} \tilde{\nabla} \times (\tilde{\tilde{U}} \times \tilde{\tilde{B}})$$

$$\implies \frac{\partial \tilde{\tilde{B}}}{\partial \tilde{t}} = \tilde{\nabla} \times (\tilde{\tilde{U}} \times \tilde{\tilde{B}})$$

So finally we have the normalized set of equations

$$\begin{cases} \frac{\partial \tilde{\tilde{U}}}{\partial \tilde{t}} + \tilde{\tilde{U}} \cdot \tilde{\nabla} \tilde{\tilde{U}} = -\tilde{\nabla} \tilde{P} + (\tilde{\nabla} \times \tilde{\tilde{B}}) \times \tilde{\tilde{B}} \\ \frac{\partial \tilde{\tilde{B}}}{\partial \tilde{t}} = \tilde{\nabla} \times (\tilde{\tilde{U}} \times \tilde{\tilde{B}}) \end{cases}$$

Now, to simplify the notation, we remove the tilde from every quantity ($\tilde{f} \rightarrow f$)

We are in Alfvénic units

$$\begin{cases} \frac{\partial \bar{U}}{\partial t} + \bar{U} \cdot \nabla \bar{U} = -\nabla P + (\nabla \times \bar{B}) \times \bar{B} \\ \frac{\partial \bar{B}}{\partial t} = \nabla \times (\bar{U} \times \bar{B}) \end{cases}$$

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